

ALPACA AI TRADING HACKATHON

SPY WEEKLY IRON BUTTERFLY

FULLY AUTONOMOUS LIVE-TRADING AGENT

Team AI Velocity Trading

Data-driven entries and exits, executed through Alpaca

Made with GAMMA

A Proven Strategy, Automated

STRATEGY BLUEPRINT

IRON BUTTERFLY



A market-neutral options structure designed to stay centered around SPY through weekly expiration.

BUY PUT

ATM - \$10

SELL PUT

ATM

SELL CALL

ATM

BUY CALL

ATM + \$10

\$10 WING WIDTH

Defined risk boundaries

MARKET-NEUTRAL STRUCTURE

Centered for range-bound conditions

5 YEARS OF HISTORICAL BACKTESTING

Validated across market regimes

REPEATABLE EXECUTION

Define structure.
Defined risk.

The Intelligence Is in the Timing

It doesn't blindly trade the clock.

How the agent decides when to enter

1 01 SEARCH BEGINS

Monday · 9:35 ET

2 02 POLL SPY MINUTE BY MINUTE

Measure the trailing 20-minute price range

3 03 WAIT FOR VOLATILITY TO SETTLE

Watch for the trailing range to stop making new lows

4 04 ENTER

After 3 minutes with no new range low

No trigger by 10:30 ET → hard timeout → enter

The agent waits for evidence that morning volatility has stabilized before committing.



From Backtest to Live Decision-Making

📄 BACKTEST

Can evaluate the entire completed historical entry window.

Can identify the quietest minute with hindsight.

📄 LIVE AGENT

Only knows what the market has revealed so far.

Looks for evidence volatility has stabilized.

Commits after 3 minutes without a new trailing-range low.

At 10:30 ET, a hard timeout guarantees execution if no earlier trigger occurs.



■ LIVE MARKET FEED // DATA_SYNC_OK

Same strategy. Live-safe decision logic.

Autonomous Risk Management

Once it's in, it keeps watching.

01 WING BREACH

Immediate close

02 +90% MAX-PROFIT TARGET

Automatically lock in gains

03 -80% MAX-LOSS STOP

Automatically close at the loss
threshold

THURSDAY - 3:40 PM ET

MANDATORY FALLBACK EXIT

Close before Friday expiration if no earlier exit condition fires.

The agent doesn't recommend an exit. It executes it.

Built on Alpaca



Every Decision Leaves a Trail

Fully auditable by design.

Every step is recorded for review and analysis.

- **DECISION** — Why the agent entered or exited
- **EXPECTED VALUE** — Pre-trade calculation
- **ORDERS & FILLS** — Execution record
- **MARKET STATE** — Price and position monitoring
- **CSV OUTPUT** — Complete post-trade history

 Every decision.
Every EV calculation.
Every fill.
Logged for post-trade analysis.

SYSTEM_LOG // AUDIT_READY

FULLY AUDITABLE

TRADE EXECUTION LOG

DECISION AUDIT

ID	ASSET	SIDE	EXPECTED VALUE	FILL PRICE	STATUS
#8421	SPY 510P	SELL	+1.24%	3.55	EXECUTED
#8422	SPY 500P	BUY	+0.88%	2.10	EXECUTED
#8423	SPY 510C	SELL	+2.15%	3.65	EXECUTED
#8424	SPY 520C	BUY	+0.45%	1.85	EXECUTED
#8425	SPY 510-520-500	IRON BFLY	+1.02%	1.25	EXECUTED

VALIDATED BEFORE DEPLOYMENT

5 YEARS OF HISTORICAL TESTING

+142.8%

Cumulative backtested return

5

YEARS

Historical test period

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DEFINED-RISK

Risk bounded by the option structure and
systematic exit rules

Historical backtest results are simulated and do not represent live trading performance.

From Strategy to Autonomous Trader

RESEARCHED STRATEGY

SPY Weekly Iron Butterfly

DYNAMIC DECISION LOGIC

Wait for volatility to settle

AUTONOMOUS EXECUTION

Enter · Monitor · Protect · Exit

ALPACA

Market Data + Trading Infrastructure

DISCIPLINED. / SYSTEMATIC. / AUTONOMOUS.

A backtested options strategy transformed into an autonomous live trader.

Alpaca AI Trading Agents Hackathon

Team AI Velocity Trading